

Started on	Sunday, 12 February 2023, 8:21 PM
State	Finished
Completed on	Sunday, 12 February 2023, 8:21 PM
Time taken	11 secs
Marks	0.00/189.00
Grade	0.00 out of 100.00



Question 1

Not answered

Marked out of 4.00

Diagnosis in the elderly

For each of the following clinical conditions, match the most appropriate psychiatric syndrome from the given list:

A 77-year-old widowed gentleman became increasingly isolated over the last two years and has been neglecting himself of late. He lives in filthy, poor conditions which has triggered a referral to the local council and social services, but he is resistant to accept help.

Choose...

A 78-year-old lady with hearing impairment lives on her own believes that her neighbours are spying on her. She can also hear neighbours talking about her all the time through the walls.

Choose...

A 83-year-old lady with bilateral cataracts sees animals and people in the house. She is relatively independent in terms of her daily activities and enjoys life in general.

Choose...

A 87-year-old lady with a history of falls was brought to the A&E by the paramedics. She is agitated, paranoid, confused and also has visual hallucinations. She was previously reported to be a high functioning lady

Choose...

Your answer is incorrect.

Explanation:

Case 1 is Diogenes syndrome (or 'senile squalor syndrome') refers to severe self-neglect seen in somewhat eccentric and reclusive elderly individuals. They become increasingly isolated and neglect themselves, living in filthy, poor conditions. It can be viewed as the response of someone with a particular personality type to the hardships of old age and loneliness (Howard & Bergman 1993). They are often oblivious to their condition and resistant to help, making management notoriously difficult as it is very hard to form a therapeutic alliance.

Case 2 is late onset paraphrenia. Persecutory delusions are the most common symptoms of late paraphrenia; they are found in around 90% of patients. Auditory hallucinations occur in approximately 75% of cases. Visual hallucinations are observed in up to 60% of patients. Few patients may present with delusions only (10-20%). Partition delusions (attack through the wall, neighbours spying via any 'partition') are common. According to ICD-10, patients must either be diagnosed as having delusional disorder or schizophrenia - no separate diagnosis exists for paraphrenia.

Case 3 is a typical presentation of Charles Bonnet syndrome.

Case 4 is delirium. The salient features include rapid onset of clinical features with fluctuations in severity over minutes and hours (even back to apparent normality), clouding of consciousness with reduced attention span and distractibility, global impairment of cognition with disorientation, and impairment of recent memory and abstract thinking.

The correct answer is: A 77-year-old widowed gentleman became increasingly isolated over the last two years and has been neglecting himself of late. He lives in filthy, poor conditions which has triggered a referral to the local council and social services, but he is resistant to accept help. → Diogenes syndrome, A 78-year-old lady with hearing impairment lives on her own believes that her neighbours are spying on her. She can also hear neighbours talking about her all the time through the walls. → Late onset paraphrenia, A 83-year-old lady with bilateral cataracts sees animals and people in the house. She is relatively independent in terms of her daily activities and enjoys life in general. → Charles Bonnet syndrome, A 87-year-old lady with a history of falls was brought to the A&E by the paramedics. She is agitated, paranoid, confused and also has visual hallucinations. She was previously reported to be a high functioning lady → Acute confusional state



Question 2

Not answered

Marked out of 5.00

Rating scales in old age psychiatry

Identify the best fitting rating scales that could be used in each of the following situations.

This scale is more useful for the elderly as it focuses on cognitive and subjective symptoms more than physical symptoms

Choose...

This scale is helpful to detect depression in people with dementia

Choose...

This scale is especially useful for older hearing-impaired subjects

Choose...

This scale is helpful for screening people with dementia

Choose...

This scale has a number of somatic items, which renders it less appropriate for older subjects.

Choose...

Your answer is incorrect.

Explanation:

- 1) Geriatric Depression Scale: Originally a 30-item test, it now has 15-items that enable most subjects to be scored for depression in four or five minutes. It avoids somatic questions and so remains useful for older patients. An overall score of five or more suggests the possibility of a depressive episode.
- 2) Cornell scale for depression is the best validated scale for detecting depression in dementing patients. It is an interviewer-administered scale that uses information both from the patient and an outside informant.
- 3) BASDEC (Brief assessment schedule depression cards): It was initially designed for use in liaison psychiatry and is particularly useful with deaf subjects. It consists of a series of statements in large print on cards, which are shown to the patient, one at a time and answers 'true' or 'false'.

4) The Mini-Mental State Examination (MMSE) is the standard screening instrument for dementia. It takes 5-10 minutes to administer and has a median positive Likelihood Ratio of 6.3 and a median Negative Likelihood Ratio of 0.19. It is a brief tool for grading cognitive impairment in elderly and screening for dementia. It is not very sensitive to change but used in the clinical trials for dementia.

5) It is a 30-point scale. It does not pick up frontal executive defects. Hamilton rating scale: It has a number of somatic items, which render it less appropriate for older subjects. It is more useful for working-age adults in quantifying depression.

The correct answer is: This scale is more useful for the elderly as it focuses on cognitive and subjective symptoms more than physical symptoms → Geriatric depression scale, This scale is helpful to detect depression in people with dementia → Cornell scale for depression, This scale is especially useful for older hearing-impaired subjects → BASDEC, This scale is helpful for screening people with dementia → MMSE, This scale has a number of somatic items, which renders it less appropriate for older subjects. → Hamilton Rating Scale



Question 3

Not answered

Marked out of 4.00

Alzheimer's disease and neuroimaging

Identify the most likely findings seen in Alzheimer's disease

CT scan	Choose... 
SPECT Scan	Choose... 
PET Scan	Choose... 
Magnetic resonance spectroscopy	Choose... 

Your answer is incorrect.

Explanation:

CT scan shows cortical atrophy especially over parietal and temporal lobes, and dilatation of the third ventricle, which correlates with cognitive impairment (most common picture). In the advanced stage of the disease, the frontal lobe is affected.

SPECT measures cerebral blood flow and in Alzheimer's disease, there is a characteristic reduction in blood flow in temporal and parietal regions. SPECT could distinguish specific features of dementia of Alzheimer type at early stages and frontal lobe dementia.

PET scan measures oxygen and glucose metabolism directly. In Alzheimer's disease, there is 20-30% reduction in oxygen and glucose metabolism in temporal and posterior parietal lobes.

In Magnetic resonance spectroscopy, there is an abnormal synthesis of membrane phospholipids early in the disease and reduced levels of N-Acetyl aspartate.

The correct answer is: CT scan → Cortical atrophy in parietal and temporal lobes, SPECT Scan → Reduced cerebral blood flow in the temporal and posterior parietal lobes, PET Scan → Reduction in oxygen and glucose metabolism in temporal and posterior parietal lobes, Magnetic resonance spectroscopy → Reduced levels of N-Acetyl aspartate

Question 4

Not answered

Marked out of 4.00

Pharmacology in the elderly

Choose ONE most appropriate medication to be prescribed for the following patients

A 74-year-old lady with a long-standing history of paranoid schizophrenia agrees to take her antipsychotic medication. She has type 2 diabetes mellitus but shows no cognitive impairment. However, the community nurse has observed a poor compliance and continuing psychosis. When family members intervened, the issue led to a bitter quarrel.

Choose... ▾

A 70-year-old gentleman with a history of chronic psychosis was started on oral flupenthixol. He suffered severe extrapyramidal side effects on taking this medication and were not keen to continue it. He has been tried already on haloperidol, quetiapine and olanzapine with no success.

Choose... ▾

A 77-year-old gentleman was recently diagnosed with Parkinson's disease. He reports having snakes in his bathroom and foxes in the living room and is extremely distressed. He is not on anti-Parkinsonian medication and is antipsychotic naive.

Choose... ▾

A 67-year-old woman suffers from low mood, anhedonia, and early morning awakening with weight loss. She is feeling isolated as her husband died a year ago.

Choose... ▾



Your answer is incorrect.

Explanation:

- 1) For patients with a history of poor compliance with oral medications, depot antipsychotics could be considered.
- 2) Offer clozapine to patients with schizophrenia whose illness has not responded adequately to treatment despite the sequential use of at least two different antipsychotic drugs, one of which should be a second-generation antipsychotic drug.
- 3) In Parkinson's disease psychosis, try an atypical antipsychotic. Low dose quetiapine is the best tolerated although extrapyramidal symptoms and stereotypical movements can occur. It is, therefore, reasonable to try quetiapine before clozapine.
- 4) For old age depression, SSRIs such as sertraline should be tried as the first line management, due to reduced side effects and relative safety in overdose.

The correct answer is: A 74-year-old lady with a long-standing history of paranoid schizophrenia agrees to take her antipsychotic medication. She has type 2 diabetes mellitus but shows no cognitive impairment. However, the community nurse has observed a poor compliance and continuing psychosis. When family members intervened, the issue led to a bitter quarrel. → Flupenthixol depot, A 70-year-old gentleman with a history of chronic psychosis was started on oral flupenthixol. He suffered severe extrapyramidal side effects on taking this medication and were not keen to continue it. He has been tried already on haloperidol, quetiapine and olanzapine with no success. → Clozapine, A 77-year-old gentleman was recently diagnosed with Parkinson's disease. He reports having snakes in his bathroom and foxes in the living room and is extremely distressed. He is not on anti-Parkinsonian medication and is antipsychotic naive. → Quetiapine, A 67-year-old woman suffers from low mood, anhedonia, and early morning awakening with weight loss. She is feeling isolated as her husband died a year ago. → Sertraline



Question 5

Not answered

Marked out of 3.00

Adverse effects of antidementia drugs

Identify the side effects caused by anti-dementia drugs for each of the following descriptions

A predictable side effect of cholinesterase inhibitors that is related to excessive cholinergic stimulation

Choose...



Commonest effect on the cardiovascular system

Choose...



Most frequent gastrointestinal effect from the commonly prescribed antidementia drugs

Choose...



Your answer is incorrect.

Explanation:

Insomnia is a mostly predictable side effect of cholinesterase inhibitors due to excessive cholinergic stimulation. Other effects include nausea, vomiting, dizziness and diarrhoea.

AchE-inhibitors may have vagotonic effects on heart rate causing bradycardia. The potential for this action may be of particular importance in patients with sick sinus syndrome or other supraventricular cardiac conduction disturbances such as a sinoatrial or atrioventricular block.

Nausea is the most common gastrointestinal side effect reported with all three drugs. Others include vomiting, diarrhoea, dyspepsia, abdominal pain, appetite disturbances.

The correct answer is: A predictable side effect of cholinesterase inhibitors that is related to excessive cholinergic stimulation → Insomnia, Commonest effect on the cardiovascular system → Bradycardia, Most frequent gastrointestinal effect from the commonly prescribed antidementia drugs → Nausea



Question 6

Not answered

Marked out of 4.00

Subtypes of dementia (1)

For each of the following descriptions choose the most appropriate diagnosis from the list shown:

A 70-year-old gentleman has a history of hypertension, recurrent TIAs, has a three-year history of step-wise deterioration in cognitive functions. On examination, he has spastic weakness of left lower limbs and an extensor plantar response. He retains good insight and there is no impairment of personality.

Choose...

A 65-year-old gentleman has a ten months history of cognitive impairment, fluctuating levels of consciousness and prominent visual hallucination. He developed severe tremor and rigidity on small dose of Haloperidol

Choose...

46-year-old gentleman presents with a history of worsening memory. On examination, there is increased tone with rigidity and abnormal jerky movements were present. There is also a strong family history of dementia.

Choose...

A 67-year-old gentleman presenting with memory problems and urinary incontinence. On neurological examination, he has a broad-based, short stepped gait. A CT brain scan shows ventricular enlargement.

Choose...

Your answer is incorrect.

Explanation:

- 1) Multi-infarct dementia is a type of vascular dementia characterised by a history of recurrent strokes/TIAs, step-wise deterioration in cognitive functions. Often patients present with positive neurological signs.
- 2) Lewy body dementia presents as progressive cognitive decline interfering with normal social or occupational function along with fluctuating cognition with profound variations in attention and alertness, recurrent visual hallucinations that are typically well-formed and detailed, and spontaneous motor features of parkinsonism. Other symptoms supporting the diagnosis include repeated falls, syncope, and transient loss of consciousness, neuroleptic sensitivity, systematised delusions and hallucinations in other modalities.

3) Huntington's disease is a progressive neurodegenerative disorder with chorea and dystonia, incoordination, cognitive decline, and behavioural difficulties. The onset of symptoms is often in middle-age after affected individuals have had children.

4) Normal Pressure Hydrocephalus is a syndrome where there is dilatation of cerebral ventricles (especially 3rd ventricle) and normal CSF pressure at lumbar puncture. It typically presents with the triad of dementia, gait ataxia, and urinary incontinence.

The correct answer is: A 70-year-old gentleman has a history of hypertension, recurrent TIAs, has a three-year history of step-wise deterioration in cognitive functions. On examination, he has spastic weakness of left lower limbs and an extensor plantar response. He retains good insight and there is no impairment of personality. → Multi-infarct dementia, A 65-year-old gentleman has a ten months history of cognitive impairment, fluctuating levels of consciousness and prominent visual hallucination. He developed severe tremor and rigidity on small dose of Haloperidol → Lewy body Dementia, 46-year-old gentleman presents with a history of worsening memory. On examination, there is increased tone with rigidity and abnormal jerky movements were present. There is also a strong family history of dementia. → Dementia in Huntington's disease, A 67-year-old gentleman presenting with memory problems and urinary incontinence. On neurological examination, he has a broad-based, short stepped gait. A CT brain scan shows ventricular enlargement. → Normal pressure hydrocephalus



Question 7

Not answered

Marked out of 4.00

Differential diagnosis of dementia

For each of the following descriptions choose the most appropriate diagnosis from the list above

A 68-year-old gentleman is presenting with memory problems, history of recurrent falls, depressed mood and tremors involving both his upper limbs for the last three years. On examination, there is marked psychomotor retardation and has cogwheel rigidity

Choose...

A 69-year-old woman is presenting with flattened mood and is becoming increasingly withdrawn. Her memory is reported to be poor. While attempting to do MMSE, she was uncooperative and frequently responds saying 'I don't know '.

Choose...

A 64 year-old-gentleman is presenting with memory loss and disinhibited behaviour. CT brain shows knife blade atrophy of frontal and temporal lobes.

Choose...

An 81-year-old lady admitted to the medical ward with a history of headache, drowsy and altered levels of consciousness. She has a history of previous hospitalisation with recurrent episodes of falls. The family has noticed physical slowness and sluggishness and also gave an account of deterioration in her cognitive functioning over the last few months

Choose...

Your answer is incorrect.

Explanation:

1) This is a case of Parkinson's disease dementia. If both motor symptoms and cognitive symptoms develop within 12 months, then it is conventional to give a diagnosis of Lewy body dementia. If the parkinsonian symptoms have existed for more than 12 months before dementia develops then a diagnosis of Parkinson's disease dementia is given.

2) Depression in elderly patients may present as dementia clinically, which is known as depressive pseudo-dementia. These patients themselves complain of memory problems, difficulty in sustaining attention & concentration and reduced intellectual capacity.

3) Dementia in Pick's disease has an insidious onset and gradual progression, early loss of personal and social awareness, early emotional blunting and an early loss of insight. CT brain scan shows bilateral asymmetrical atrophy of frontal and temporal lobes, also called as knife-blade atrophy.

4) Chronic subdural haematoma is relatively more common in old age. In older people, as the brain shrinks the veins, which lie in between it and skull, can get torn easily even after minor trauma. SDH should be borne in mind where there is a changing pattern in cognitive function, especially if risk factors for SDH exist. The important risk factors are post trauma; elderly after a fall; elderly after a head injury, infancy, cerebral atrophy, alcoholism, epilepsy, clotting disorders, pre-disposing drugs such as aspirin, warfarin, etc.

The correct answer is: A 68-year-old gentleman is presenting with memory problems, history of recurrent falls, depressed mood and tremors involving both his upper limbs for the last three years. On examination, there is marked psychomotor retardation and has cogwheel rigidity → Dementia in Parkinson's disease., A 69-year-old woman is presenting with flattened mood and is becoming increasingly withdrawn. Her memory is reported to be poor. While attempting to do MMSE, she was uncooperative and frequently responds saying 'I don't know '. → Depressive pseudodementia, A 64 year-old-gentleman is presenting with memory loss and disinhibited behaviour. CT brain shows knife blade atrophy of frontal and temporal lobes. → Dementia in Pick's disease, An 81-year-old lady admitted to the medical ward with a history of headache, drowsy and altered levels of consciousness. She has a history of previous hospitalisation with recurrent episodes of falls. The family has noticed physical slowness and sluggishness and also gave an account of deterioration in her cognitive functioning over the last few months → Chronic subdural haematoma



Question 8

Not answered

Marked out of 4.00

Subtypes of dementia (2)

Identify the type of dementia using the descriptions given below:

This is subcortical dementia with a clinical course characterised by gradual intellectual decline, generalised slowing, and motor problems. MRI scan shows multiple microvascular infarcts and white matter leukoariosis

Choose...

The disease in humans affects mainly young men in their 20s and is characterised by early anxiety and depressive symptoms, followed by personality changes, and finally progressive dementia. Ataxia and myoclonus are prominent.

Choose...

This condition typically presents with the triad of dementia, gait ataxia, and urinary incontinence.

Choose...

The disease usually presents as a progressive frontal dementia. Classic symptoms would include grandiosity, euphoria, and mania with mood-congruent delusions. Disinhibition, personality change, and memory impairment are also common.

Choose...

Your answer is incorrect.

Explanation:

Case 1 is Binswanger's disease (progressive small vessel disease). It is subcortical dementia with a clinical course characterised by slow intellectual decline and generalised slowing. MRI scan shows small distinct infarcts (lacunae) or more widespread white matter changes (leukoariosis).

Case 2 is new variant CJD. The rise of vCJD followed an epidemic of bovine spongiform encephalopathy (BSE) in cattle. The disease in humans affects mainly young men in their 20s and is characterised by early anxiety and depressive symptoms, followed by personality changes, and finally progressive dementia. Ataxia and myoclonus are prominent and the typical course is 1-2 yrs until death.

Case 3 is Normal pressure hydrocephalus. NPH is a syndrome where there is dilatation of cerebral ventricles, especially the third ventricle) and normal CSF pressure at lumbar puncture. It typically presents with the triad of dementia, gait ataxia, and urinary incontinence.

Case 4 is General Paresis of insane (GPI). It is an impairment of mental function caused by damage to the brain from untreated syphilis. This damage can produce hallucinations and delusions (grandiosity). Brief, sharp "lightning" pains occur with tabes dorsalis. Eye changes including abnormal pupil response, mood changes (euphoria and mania), overactive reflexes, personality changes and speech changes are also noted. General paresis usually begins about 15 - 20 years after the syphilis infection. Cerebral atrophy and gliosis, dominantly in the frontal lobes are well-known neuropathological findings of GPI.

The correct answer is: This is subcortical dementia with a clinical course characterised by gradual intellectual decline, generalised slowing, and motor problems. MRI scan shows multiple microvascular infarcts and white matter leukoariosis → Binswanger's disease, The disease in humans affects mainly young men in their 20s and is characterised by early anxiety and depressive symptoms, followed by personality changes, and finally progressive dementia. Ataxia and myoclonus are prominent. → New variant CJD, This condition typically presents with the triad of dementia, gait ataxia, and urinary incontinence. → Normal pressure hydrocephalus, The disease usually presents as a progressive frontal dementia. Classic symptoms would include grandiosity, euphoria, and mania with mood-congruent delusions. Disinhibition, personality change, and memory impairment are also common. → General paresis of insane



Question 9

Not answered

Marked out of 4.00

Psychiatric side effects of non-psychiatric medications

For each of the side effect profile given below identify the possible offending drug

A 74-year-old gentleman being treated for resting tremors develops acute confusional state with paranoia and visual hallucinations

A 67-year-old lady was taking treatment for swollen legs complaints of visual halos, which are green in colour and appears to be confused.

A 56-year-old woman on treatment for polymyalgia rheumatica becomes hypomanic and confused.

A 79-year-old man treated for ventricular tachycardia became confused, agitated and presented with paranoid delusions

Your answer is incorrect.

Explanation:

The patient in scenario 1 is likely to be taking treatment for Parkinson's disease using levodopa. Levodopa is associated with high rates of visual hallucinations. Other psychiatric side effects of Levodopa consist of psychosis, hallucinations, confusion, euphoria, depression, insomnia and abnormal dreams.

The patient in scenario 2 is likely to be suffering from cardiac failure and is taking digoxin. One of the side effects of cardiac glycosides such as digoxin is a change in vision characterised by seeing a halo or light around objects.

The patient in scenario 3 is likely to be taking steroids for her autoimmune condition. Corticosteroids are commonly implicated in iatrogenic depression, mania or hypomania and confusion.

Amiodarone is used for the treatment of refractory ventricular tachycardia (scenario 4). The psychiatric side effects include altered mental status, hallucinations, and delirium that have been reported mainly among older subjects.

The correct answer is: A 74-year-old gentleman being treated for resting tremors develops acute confusional state with paranoia and visual hallucinations → Levodopa, A 67-year-old lady was taking treatment for swollen legs complaints of visual halos, which are green in colour

and appears to be confused. → Digoxin, A 56-year-old woman on treatment for polymyalgia rheumatica becomes hypomanic and confused.
→ Prednisolone, A 79-year-old man treated for ventricular tachycardia became confused, agitated and presented with paranoid delusions → Amiodarone



Question 10

Not answered

Marked out of 4.00

Treatment options of reversible dementia (1)

For each clinical situation given below, which of the listed treatments may provoke a partial or fully recoverable state?

A 60-year-old lady presents with recent onset memory loss and early morning headache and nausea.

Choose... ▾

A 45-year-old man with long-standing alcohol problem presents with fluctuating alertness and asterixis.

Choose... ▾

A 60-year-old man presents with gait ataxia, urinary incontinence and mild cognitive impairment.

Choose... ▾

A 67-year-old gentleman presents with grandiose delusions, irritability, memory loss and abnormal pupils

Choose... ▾

Your answer is incorrect.

Explanation:

1) This patient has a cerebral tumour which causes symptoms common to all space occupying lesions and consequent upon raised intracranial pressure (headache, early morning nausea, vomiting, impaired cognition, apathy, emotional blunting, altered consciousness and brain herniation syndromes). Some of these compression symptoms could be treated with corticosteroids such as dexamethasone, which can reduce cerebral oedema and relieve pressure symptoms. However, surgical evacuation would be the best treatment of choice for patients with cerebral tumour.

2) This patient has developed hepatic failure due to chronic alcoholism, a fluctuating chronic organic state in which physical and psychiatric symptoms wax and wane in unison. The physical symptoms include flapping tremor of hands (asterixis), motor retardation, ataxia, hyperreflexia, clonus & coma. Neomycin and lactulose are used for the treatments of hepatic encephalopathy.

3) This is a case of normal pressure hydrocephalus, featured by dementia, urinary incontinence and gait ataxia. Treatment most commonly requires the insertion of a shunt between the lateral ventricles and the inferior vena cavae and this may alleviate even severe degrees of dementia (Bekkelund et al 1999).

4) The diagnosis is general paresis of insane (GPI) due to neurosyphilis, caused by *Treponema pallidum*. Antibiotic therapy prevents disease progression. Neurosyphilis is treated with intramuscular procaine penicillin for 10-21 days.

The correct answer is: A 60-year-old lady presents with recent onset memory loss and early morning headache and nausea. → Dexamethasone, A 45-year-old man with long-standing alcohol problem presents with fluctuating alertness and asterixis. → Neomycin and lactulose, A 60-year-old man presents with gait ataxia, urinary incontinence and mild cognitive impairment. → Surgical shunt, A 67-year-old gentleman presents with grandiose delusions, irritability, memory loss and abnormal pupils → Antibiotic therapy



Question 11

Not answered

Marked out of 4.00

Treatment options of reversible dementia (2)

For each clinical situation given below, which of the listed treatments may result in a partial or fully recoverable state?

A 45-year-old chronic alcoholic now presents with apathy, fatigue, irritability, depression and cognitive impairment.

Choose... ▾

A 76-year-old woman with atrophic gastritis now presents with fatigue, tingling and numbness of hands & feet, depression and memory problems.

Choose... ▾

A 57-year-old woman presents with mouth ulcers, diarrhoea, lassitude, impaired cognition, depression and dementia

Choose... ▾

A 45-year-old woman with lethargy, excessive tiredness, constipation, low mood and memory problems.

Choose... ▾

Your answer is incorrect.

Explanation:

1) The diagnosis is chronic thiamine deficiency due to alcoholism. Acute thiamine depletion causes Wernicke's encephalopathy and later, Korsakoff's psychosis. Chronic thiamine deficiency causes apathy, fatigue, irritability, depression and cognitive impairment prior to the development of classical beriberi.

2) The diagnosis is vitamin B 12 deficiency (Pernicious anemia). The typical symptoms of vitamin B12 deficiency anemia include diarrhoea or constipation, fatigue, loss of appetite, pale skin, problems concentrating, swollen, red tongue or bleeding gums. Monthly vitamin B12 injections are prescribed to correct the vitamin B12 deficiency.

3) The diagnosis is folate deficiency. It may cause diarrhoea, grey hair, mouth ulcers and peptic ulcers. It may be associated with lassitude, impaired cognition, depression and dementia (prolonged deficiency). The treatment would be replacement with folic acid supplements and fortified foods, to ensure the proper daily intake.

4) This is hypothyroidism, caused by low thyroxine levels due to thyroid disease, thyroidectomy or drugs. Mental symptoms develop insidiously and may include impaired concentration, psychomotor retardation, depression and dementia and 'myxedematous madness', a form of paranoid psychosis with delusions and hallucinations. The psychiatric symptoms usually resolve with thyroxine replacement therapy.

The correct answer is: A 45-year-old chronic alcoholic now presents with apathy, fatigue, irritability, depression and cognitive impairment. → Thiamine, A 76-year-old woman with atrophic gastritis now presents with fatigue, tingling and numbness of hands & feet, depression and memory problems. → Vitamin B12, A 57-year-old woman presents with mouth ulcers, diarrhoea, lassitude, impaired cognition, depression and dementia → Folate supplements, A 45-year-old woman with lethargy, excessive tiredness, constipation, low mood and memory problems. → Thyroxine



Question 12

Not answered

Marked out of 3.00

Early onset dementia (1)

For each of the following situations, identify the correct diagnosis from the list above

A 61-year-old man presents with paralysis of the extraocular muscles, cervical dystonia, dementia and ataxia.

Choose... ▾

A 53-year-old man with clumsiness, uncontrollable twitching of face, shoulders, cognitive impairment and ataxia. There is a strong family history.

Choose... ▾

A 25-year-old man with jaundice, wing-beating tremor of shoulders, rigidity, and memory problems

Choose... ▾

Your answer is incorrect.

Explanation:

1) The diagnosis is Progressive Supranuclear Palsy. The onset is in the sixth decade (range 45 to 75 years]. It presents with difficulty in balance, abrupt falls, slurred speech, dysphagia, and vague changes in personality, sometimes with an apprehensiveness and fretfulness suggestive of an agitated depression. The most common early complaint is unsteadiness of gait and unexplained falling (retropulsion).

2) Huntington's disease is a progressive neurodegenerative disorder with chorea and dystonia, incoordination, cognitive decline, and behavioural difficulties. The onset of symptoms is often in middle-age after affected individuals have had children.

3) The diagnosis is Wilson's disease. The onset is usually in adolescence or early childhood. Clinical features include liver cirrhosis, tremor, dystonia, flapping tremor of wrists, wing-beating tremor of shoulders, and Kayser-Fleischer rings (greenish brown corneal deposits). Mood disturbances are common and subcortical dementia is seen in 25% of cases.

The correct answer is: A 61-year-old man presents with paralysis of the extraocular muscles, cervical dystonia, dementia and ataxia. → Progressive supranuclear palsy, A 53-year-old man with clumsiness, uncontrollable twitching of face, shoulders, cognitive impairment and ataxia. There is a strong family history. → Huntington's disease, A 25-year-old man with jaundice, wing-beating tremor of shoulders, rigidity, and memory problems → Wilson's disease

Question 13

Not answered

Marked out of 3.00

Early onset dementia (2)

For each of the following situations, identify the correct diagnosis from the list above

A 43-year-old man with lymphoma now presents with clumsiness, cognitive impairment and problems with co-ordination

Choose...

A 56-year-old man with symptoms of Parkinson's disease has developed dementia associated with fluctuating cognition and sensitivity to antipsychotic medications

Choose...

A 59-year-old man is presenting with rapidly progressive dementia, myoclonus and ataxia.

Choose...

Your answer is incorrect.

Explanation:

1) Progressive multifocal leukoencephalopathy (PML) is caused by the reactivation of a common virus in the central nervous system of immune-compromised individuals. PML is usually seen in people with AIDS, and those undergoing chronic corticosteroid or immunosuppressive therapy and individuals with cancer, such as Hodgkin's disease, lymphoma, and sarcoidosis. The most prominent symptoms are clumsiness; progressive weakness; cognitive impairment, problems with coordination and visual impairment, speech disturbances and sometimes personality changes. The progression of deficits leads to life-threatening disability and death over weeks to months. (Ref: www.ninds.nih.gov).

2) Patients with Lewy body dementia have a history of Parkinsonism and are very sensitive to antipsychotic medications.

3) In sporadic CJD, most cases occur after the fifth decade, although onset can occur at any age. The clinical picture is one of rapidly deteriorating dementia, myoclonus, cerebellar and extrapyramidal signs leading to death within a year. Patients may present with non-specific symptoms such as lethargy, depression and fatigue. Within weeks, more fulminant symptoms develop, including progressive cortical-pattern dementia, myoclonus and pyramidal and extrapyramidal signs. Myoclonus becomes prominent as the disease progresses. Patients may develop cortical blindness.

The correct answer is: A 43-year-old man with lymphoma now presents with clumsiness, cognitive impairment and problems with co-ordination → Progressive multifocal leucoencephalopathy, A 56-year-old man with symptoms of Parkinson's disease has developed dementia associated with fluctuating cognition and sensitivity to antipsychotic medications → Lewy body dementia, A 59-year-old man is presenting with rapidly progressive dementia, myoclonus and ataxia. → Creutzfeldt-Jakob disease



Question 14

Not answered

Marked out of 3.00

Hoarding in the elderly

Which one of the listed options best describes each of the following clinical situations?

A 60-year-old man who is known to mental health services for a long time appears to be self-neglecting and is collecting various unusual things.

Choose...

An elderly lady has started to collect things many years ago with the fear that she might lose them if not stored in an orderly fashion. When her daughter tried to get rid of these things but she became severely anxious and upset with the fear that she would lose all her belongings.

Choose...

A 60-year-old woman shows a gradual change in personality characterised by increasing disinhibition and forgetfulness. She has collected multiple items over the course of last few years. There is a family history of memory problems.

Choose...

Your answer is incorrect.

Explanation:

- 1) Diogenes or senile squalor syndrome is usually seen in patients with decomposing personality disorders.
- 2) This type of hoarding associated with anxiety is suggestive of the obsessive-compulsive disorder.
- 3) The presentation is suggestive of Pick's disease where a range of repetitive behaviors can occur. Repetitive behaviors encompass simple repetitive acts and verbal or motor stereotypies or complex repetitive motor routines such as counting, checking, cleaning, wandering along a fixed route, repeated trips to the bathroom, collecting & hoarding objects, pathological gambling, and rituals involving touching, grabbing, and superstitious acts.

The correct answer is: A 60-year-old man who is known to mental health services for a long time appears to be self-neglecting and is collecting various unusual things. → Diogenes syndrome, An elderly lady has started to collect things many years ago with the fear that she might lose them if not stored in an orderly fashion. When her daughter tried to get rid of these things but she became severely anxious and upset with the fear that she would lose all her belongings. → OCD, A 60-year-old woman shows a gradual change in personality

characterised by increasing disinhibition and forgetfulness. She has collected multiple items over the course of last few years. There is a family history of memory problems. → Pick's disease

Question 15

Not answered

Marked out of 5.00

Anti-dementia drug dosages

Identify the usual treatment dose for each of the following anti-dementia drugs

Donepezil

Choose...



Rivastigmine

Choose...



Rivastigmine transdermal patch

Choose...



Galantamine

Choose...



Memantine

Choose...



Your answer is incorrect.

Explanation: Donepezil-10 mg daily Rivastigmine- 6mg bd Rivastigmine patch- 9.5 mg/24 hours, Galantamine- 12 mg bd, Memantine - 10 mg bd

The correct answer is: Donepezil → 10 mg daily, Rivastigmine → 6 mg bd, Rivastigmine transdermal patch → 9.5 mg/24 hours, Galantamine → 12 mg bd, Memantine → 10 mg bd

Question 16

Not answered

Marked out of 5.00

Investigations in the elderly

identify the investigations, which would most likely confirm the diagnosis in each of the following clinical situations

A 79-year-old man is presenting with progressive short-term memory loss for last 18 months. He has difficulty in performing complex activities of daily living tasks.

Choose...

An 83-year-old man is presenting with a progressive cognitive decline, apathy and visual hallucinations. He has a history of Parkinson's disease

Choose...

A 67-year-old man presenting with memory loss, language difficulties, disinhibition and personality changes over last one year.

Choose...

A 63-year-old man with recent onset memory problems is showing a rapid declining with abnormal twitchy movements and gait disturbance.

Choose...

A 67-year-old gentleman with a history of memory problems for last six months. He has a history of hypertension and has had recurrent TIAs in the past.

Choose...

Your answer is incorrect.

Explanation:

1) Diagnosis is Alzheimer's disease. CT/MRI shows medial temporal lobe atrophy and parietal atrophy.

2) The diagnosis is Lewy body dementia. Motor retardation and parkinsonism are often reported in DLB and have been linked with striatal dopamine deficits, but the affective and cognitive features of apathy are also related to fronto-subcortical dysfunctions involving basal ganglia. DaTSCAN shows reduced striatal dopamine uptake in DLB.

3) Diagnosis is a fronto-temporal type of dementia. In this condition, SPECT scan shows decreased blood flow and perfusion in the frontal and temporal lobes of the brain.

4) The diagnosis is Creutzfeldt-Jakob disease, which is multifocal dementia, involving cerebrum and cerebellum. CT/MRI shows atrophy of both cerebrum and cerebellum.

5) The diagnosis is vascular dementia. MRI scan shows global atrophy, focal white matter lesions, lacunes and strategic infarcts (infarcts in regions that are involved in cognitive function).

The correct answer is: A 79-year-old man is presenting with progressive short-term memory loss for last 18 months. He has difficulty in performing complex activities of daily living tasks. → CT shows medial temporal lobe atrophy, An 83-year-old man is presenting with a progressive cognitive decline, apathy and visual hallucinations. He has a history of Parkinson's disease → DaTSCAN shows reduced striatal dopamine uptake, A 67-year-old man presenting with memory loss, language difficulties, disinhibition and personality changes over last one year. → HMPAO SPECT scan shows reduced perfusion in the frontal lobes, A 63-year-old man with recent onset memory problems is showing a rapid declining with abnormal twitchy movements and gait disturbance. → CT shows atrophy of cortex and cerebellum, A 67-year-old gentleman with a history of memory problems for last six months. He has a history of hypertension and has had recurrent TIAs in the past. → MRI shows focal white matter lesions



Question 17

Not answered

Marked out of 4.00

EEG Abnormalities

Identify the most likely disorders in which each of the following EEG changes are observed:

Generalized periodic 1-2 Hz sharp wave complexes

Choose...



Episodic discharges recurring every 1-3 seconds

Choose...



Flattened trace

Choose...



Triphasic waves and slowing of the normal background frequency

Choose...



Your answer is incorrect.

Explanation:

- 1) Creutzfeldt-Jakob disease- Generalised periodic 1-2 Hz sharp waves are seen in nearly 90% patients with sporadic CJD. Less often in familial / hormonal transplant related forms. NOT seen in the variant form.
- 2) Herpes simplex encephalitis- Episodic discharges are recurring every 1-3 seconds with variable focal waves over the temporal areas.
- 3) Huntington's disease- Initial loss of alpha; later flattened trace.
- 4) Hepatic encephalopathy-Diffuse generalized slowing. Triphasic waves: 1.5 to 3.0 per second high-voltage slow waves especially in hepatic encephalopathy

The correct answer is: Generalized periodic 1-2 Hz sharp wave complexes → Creutzfeldt-Jakob disease, Episodic discharges recurring every 1-3 seconds → Herpes simplex encephalitis, Flattened trace → Huntington's dementia, Triphasic waves and slowing of the normal background frequency → Hepatic encephalopathy

Question 18

Not answered

Marked out of 3.00

Prion disorders

Identify the cognitive disorder for each of the descriptions given below

Thought to have been transmitted by ritual cannibalism and reported among some tribes in the New Guinea

Choose...

The commonest prion protein dementia and consists of a dementing illness which runs a rapid course with death occurring within few months

Choose...

Clinical presentation similar to CJD but with a longer duration to death

Choose...

Your answer is incorrect.

Explanation:

1) Kuru is thought to have been transmitted by ritual cannibalism due to eating the infected brain tissue and reported among some tribes of New Guinea. It is a type of prion protein dementia and symptoms include personality changes, behavioural disturbances, cerebellar ataxia, dementia and death.

2) Description 2 fits CJD, the commonest of all prion protein dementias. The clinical picture is one of rapidly deteriorating dementia, myoclonus, cerebellar and extrapyramidal signs leading to death within a year.

3) Gerstmann-straussler syndrome is a rare prion disease with a clinical presentation similar to CJD. It is characterized by progressive cerebellar ataxia and dementia but with a longer duration to death (usually within five years of diagnosis).

The correct answer is: Thought to have been transmitted by ritual cannibalism and reported among some tribes in the New Guinea → Kuru, The commonest prion protein dementia and consists of a dementing illness which runs a rapid course with death occurring within few months → Creutzfeldt-Jakob disease, Clinical presentation similar to CJD but with a longer duration to death → Gerstmann-Straussler syndrome

Question 19

Not answered

Marked out of 4.00

Genetics of Alzheimer's disease

Identify the chromosome locus for each of the following genes implicated in Alzheimer's disease

Presenilin 1 gene

Choose... ▾

Presenilin 2 gene

Choose... ▾

Beta-amyloid precursor protein gene

Choose... ▾

Apolipoprotein e4 gene

Choose... ▾

Your answer is incorrect.

Explanation:

Presenilin 1 gene (Chromosome 14) is associated with early onset Alzheimer's disease.

Presenilin 2 genes (Chromosome 1) is also associated with early onset Alzheimer's disease.

Beta Amyloid precursor protein gene (Chromosome 21) is the gene for Amyloid precursor protein and is found on the long arm; it is associated with early onset Alzheimer's disease.

Apolipoprotein e 4 (Chromosome 19) - Presence of e 4 alleles increases the risk of late-onset dementia of Alzheimer's type.

The correct answer is: Presenilin 1 gene → Chromosome 14, Presenilin 2 gene → Chromosome 1, Beta-amyloid precursor protein gene → Chromosome 21, Apolipoprotein e4 gene → Chromosome 19

Question 20

Not answered

Marked out of 4.00

Dementia & drug management

Identify the most appropriate drug based on the following descriptions:

This is the only drug currently licensed in the UK for behavioural and psychological symptoms of dementia

Choose... ▾

This medication may increase the risk of bleeding and should be used with caution

Choose... ▾

The use of this drug was associated with increased mortality in patients with mild cognitive impairment when compared with placebo

Choose... ▾

This antedementia drug is prone to cause confusion and hallucinations in patients

Choose... ▾

Your answer is incorrect.

Explanation:

1) Risperidone is the only drug licensed in the UK for behavioural and psychological symptoms of dementia. It is indicated for the short-term treatment (up to 6 weeks) of persistent aggression in patients with moderate to severe Alzheimer's disease who are not responding to non-pharmacological measures and when there is a risk of harm to self or others.

2) Several reports have noted that the drug Gingko biloba may increase the risk of bleeding and should be used with caution.

3) The use of galantamine was associated with increased mortality in patients with mild cognitive impairment when compared with placebo. Although no specific cause was predominant, half the deaths reported were due to cardiovascular disorders. As a result, the FDA issued a warning restricting the use of galantamine in patients with mild cognitive impairment (MCI).

4) Memantine can cause confusion and hallucinations in patients although it is rare. The common side effects are headache, dizziness, constipation, somnolence and hypertension.

The correct answer is: This is the only drug currently licensed in the UK for behavioural and psychological symptoms of dementia → Risperidone, This medication may increase the risk of bleeding and should be used with caution → Gingko Biloba, The use of this drug was

associated with increased mortality in patients with mild cognitive impairment when compared with placebo → Galantamine, This antimentia drug is prone to cause confusion and hallucinations in patients → Memantine



Question 21

Not answered

Marked out of 5.00

Treatment options for dementia

Identify the most appropriate drug described in each of the following statements:

This medication may be neuroprotective and has a disease modifying effect

Choose...

This hormone may have actions that have preventive effect and potentially reduces the risk of Alzheimer's disease

Choose...

This drug could be used in mild-moderate Alzheimer's dementia if patients cannot tolerate any of the acetylcholinesterase inhibitors

Choose...

This is an acetylcholinesterase drug that allosterically modulates nicotinic receptors

Choose...

This antidementia drug has been withdrawn for routine use due to hepatotoxicity

Choose...

Your answer is incorrect.

Explanation: 1) Memantine acts as an antagonist at N-Methyl-D-Aspartate (NMDA) receptors, an action that, in theory, may be neuroprotective and thus disease - modifying drug.

2) The hormone oestrogen may have actions that have a preventive effect and potentially reduces the risk of Alzheimer's disease.

3) According to latest NICE guidelines, memantine could be used in patients with mild-moderate Alzheimer's dementia if patients cannot tolerate any of the acetylcholinesterase inhibitors.

4) Galantamine is an acetylcholinesterase drug that also allosterically modulates nicotinic receptors.

5) Tacrine is rarely used, because of its potential for hepatotoxicity.

The correct answer is: This medication may be neuroprotective and has a disease modifying effect → Memantine, This hormone may have actions that have preventive effect and potentially reduces the risk of Alzheimer's disease → Oestrogen, This drug could be used in mild-moderate Alzheimer's dementia if patients cannot tolerate any of the acetylcholinesterase inhibitors → Memantine, This is an

acetylcholinesterase drug that allosterically modulates nicotinic receptors → Galantamine, This antimentia drug has been withdrawn for routine use due to hepatotoxicity → Tacrine



Question 22

Not answered

Marked out of 3.00

Dementia

Which one of the listed options best describes each of the following clinical scenarios?

A 53-year-old man is unable to direct eye movements voluntarily but has normal reflex eye movements.

Choose...

A 37-year-old woman with fulminant liver failure and abnormal movements

Choose...

A 79-year-old gentleman with psychomotor slowness, memory loss and gradual intellectual decline

Choose...

Your answer is incorrect.

Explanation:

1) Progressive supranuclear palsy- The most common early complaint is unsteadiness of gait and unexplained falling (retropulsion). It may take a year or longer for the characteristic syndrome-comprising supranuclear ophthalmoplegia, pseudo bulbar palsy, and axial dystonia to develop fully. Difficulty in the voluntary vertical movement of the eyes (vertical gaze palsy). Bell's phenomenon (reflexive upturning of eyes on forced closure of the eyelids) and the ability to converge are also lost eventually, and the pupils then become small.

2) Wilson's disease- It is a rare autosomal recessive, progressive, but eminently treatable disorder of copper metabolism, causing a personality change, cognitive decline, and extrapyramidal disorder and cirrhosis of the liver. The extrapyramidal features include tremor, dysarthria, drooling, rigidity and bradykinesia. Kayser-Fleischer rings (copper deposits on Descemet's membrane of the cornea) are present in 95% of patients with neurological symptoms.

3) Progressive small vessel disease (Binswanger's disease)- This is subcortical dementia with a clinical course characterised by slow intellectual decline and generalised slowing. The clinical picture can be dominated by dementia, and there may be concomitant physical problems such as gait disorders or dysarthria. MRI scan shows small distinct infarcts (Lacunae) or more generalised white matter changes (Leukoariosis). Depression and Pseudobulbar palsy are not uncommon.

The correct answer is: A 53-year-old man is unable to direct eye movements voluntarily but has normal reflex eye movements. → Progressive supranuclear palsy, A 37-year-old woman with fulminant liver failure and abnormal movements → Wilson's disease, A 79-year-

old gentleman with psychomotor slowness, memory loss and gradual intellectual decline → Binswanger's disease



Question 23

Not answered

Marked out of 3.00

Dementia- subtypes

For each of the following descriptions choose the most appropriate diagnosis

A 60-year-old woman has been wandering the streets and was taken to A&E. She appears unconcerned during the assessment. She has diabetes and hypertension. CT shows a subcortical pattern of excessive white matter lesions with no other notable findings.

Choose...

A 70-year-old man with a history of recent onset cognitive decline and visual hallucinations. He has been attending the neurology clinic for the past three years, and his medication includes a list of drugs also to Selegiline.

Choose...

A 67-year-old man with a history of progressive cognitive decline, loss of visuospatial and calculation abilities and reduction in functioning.

Choose...

Your answer is incorrect.

Explanation:

1) Case 1 is Binswanger's disease (a progressive small vessel disease). This is subcortical dementia with a clinical course characterized by slow intellectual decline and generalised slowing. There is a history of vascular risk factors like diabetes and hypertension. The clinical picture may be dominated by dementia, and there may be concomitant physical problems such as gait disorders or dysarthria. MRI scan shows small distinct infarcts (lacunae) or more generalised white matter changes (Leukoariosis).

2) This patient is on Selegiline, which is a drug used for Parkinson's disease and has been reviewed by the neurology team. It is a case of Parkinson's disease dementia. If the parkinsonian symptoms have existed for more than 12 months before dementia develops then a diagnosis of Parkinson's disease dementia is given. Otherwise, you can consider Lewy Body Dementia as an option.

3) It is a case of Alzheimer's dementia featured a history of progressive cognitive decline and reduction in functioning. It is a type of cortical dementia involving temporoparietal dysfunction such as loss of visuospatial and calculation abilities and reduction in functioning.

The correct answer is: A 60-year-old woman has been wandering the streets and was taken to A&E. She appears unconcerned during the assessment. She has diabetes and hypertension. CT shows a subcortical pattern of excessive white matter lesions with no other notable

findings. → Binswanger's disease, A 70-year-old man with a history of recent onset cognitive decline and visual hallucinations. He has been attending the neurology clinic for the past three years, and his medication includes a list of drugs also to Selegiline. → Parkinson's disease dementia, A 67-year-old man with a history of progressive cognitive decline, loss of visuospatial and calculation abilities and reduction in functioning. → Alzheimer's disease



Question 24

Not answered

Marked out of 4.00

Mood Disorders In the Elderly:

Choose the correct diagnosis for the situations given below:

Mr Y is a man in his 70s', still working voluntarily four days a week. He has short-term memory loss for 3-4 months. On MMSE, he scored 27/30 and had problems with recall. His affect is described to be reactive.

Choose...

Mrs Y is a 73-year-old woman who has short-term memory loss for last eight months. On MMSE, she has problems with calculation and could not draw a pentagon on MMSE. She is not working. Her mood is described as euthymic.

Choose...

Mrs Z has low mood for last three months and is having early morning headaches & vomiting

Choose...

Mr P had a stroke ten months ago. He now presents with apathy and psychomotor retardation. He does not have any tearfulness or pessimism. His MMSE score is 26/30, and he has dysexecutive function

Choose...

Your answer is incorrect.

Explanation:

- 1) This man has only got a short term-memory loss but scored high on MMSE with a score of 27/30. There is no impairment in his daily functioning, which is more suggestive of the mild cognitive disorder.
- 2) It is a case of dementia featured by a history of progressive cognitive decline and reduction in functioning. It is a type of cortical dementia involving temporo-parietal dysfunction such as loss of visuospatial and calculation abilities and reduction in functioning.
- 3) This patient is exhibiting symptoms of an intracranial tumour- benign/malignant. 20% of brain tumour patients also present with psychiatric symptoms such as depressed mood, which is classified under organic mood disorder.
- 4) The clinical feature of vascular depression would include apathy, psychomotor retardation, poor executive function on cognitive testing, less depressive thinking such as guilt and worthlessness and late age of onset. This type is probably less responsive to antidepressant drugs (Simpson et al. 1998) but patients may recover with ECT, although with an increased risk of post-treatment delirium

The correct answer is: Mr Y is a man in his 70s', still working voluntarily four days a week. He has short-term memory loss for 3-4 months. On MMSE, he scored 27/30 and had problems with recall. His affect is described to be reactive. → Mild Cognitive Impairment, Mrs Y is a 73-year-old woman who has short-term memory loss for last eight months. On MMSE, she has problems with calculation and could not draw a pentagon on MMSE. She is not working. Her mood is described as euthymic. → Dementia, Mrs Z has low mood for last three months and is having early morning headaches & vomiting → Organic mood disorder, Mr P had a stroke ten months ago. He now presents with apathy and psychomotor retardation. He does not have any tearfulness or pessimism. His MMSE score is 26/30, and he has dysexecutive function → Vascular depression



Question 25

Not answered

Marked out of 6.00

Investigations in the elderly

For each of the following clinical conditions, match the correct investigations to be performed from the given list:

1. A 77-year-old lady is admitted to the geriatric ward with a history of an acute confusional state. The patient passed urine several times at night and it was reported to be foul smelling. (choose TWO)

2. A 79-year-old lady who lives on her own admits to drinking three glasses of wine everyday. She is admitted to the ward for depression and alcohol misuse. (choose TWO)

3. A 83-year-old lady with the history of diabetes and hypertension was admitted to the ward with a history of gradual intellectual decline, depression and cognitive impairment. She is unsteady on her feet and is also dysarthric. (choose ONE)

4. A 77-year-old lady who is on lithium augmentation for her depressive symptoms is constipated, lethargic and has gained weight despite poor appetite in the last six months.(choose ONE)

Explanation:

Case 1 is suggestive of a woman possibly suffering from a urinary tract infection. Full blood count (Leucocytosis may indicate the seriousness of the UTI) and mid-stream urine are investigations of choice.

Case 2 describes chronic alcoholism. Raised Gamma glutamyl transferase (GGT) and raised mean corpuscular volume (MCV) are important features seen in alcoholic subjects.

Case 3 describes vascular dementia (Binswanger's disease). MRI scan is the investigation of choice for patients with vascular dementia, subcortical type.

Case 4 is a patient who has developed hypothyroidism, which occurs on long-term treatment with Lithium carbonate. Abnormal thyroid function test especially raised TSH levels, is seen in patients with lithium-induced hypothyroidism.

Correct Answer is : 1. Full blood count, Midstream urine 2. Full blood count, Liver function tests 3. MRI Scan 4. Thyroid function tests



Question 26

Not answered

Marked out of 8.00

Dementia & Classifications (1)

Identify the correct classification of dementia based on the primary site of pathological changes as stated below

1. Cortical dementia (Choose three answers) ❌ ❌ ❌2. Subcortical dementia (Choose three answers) ❌ ❌ ❌3. Cortico-subcortical dementia ❌4. Multifocal dementia ❌

Explanation: 1) Dementias may be classified in terms of the primary site of pathology. Cortical dementias include those affecting frontotemporal regions such as Pick's disease, normal pressure hydrocephalus, and motor neuron disease, and those affecting posterior parietal regions such as Alzheimer's disease.

2) Subcortical dementias include Parkinson's disease dementia, Huntington's disease, HIV-associated dementia, Binswanger's disease and Wilson's disease. These are characterised by gross psychomotor slowing, abnormal movements, low mood, and apathetic personality but somewhat less prominent amnesia.

3) Cortico-sub cortical dementias include Lewy Body Dementia where both cortical and subcortical symptoms are present together.

4) Multifocal dementias include Creutzfeldt-Jakob disease. It involves not only cerebral cortex but also cerebellum and subcortical structures.

Ref: Oxford Handbook of Psychiatry 1e: Pg 133

Correct Answer is : 1. Alzheimer's disease, Pick's disease, Normal pressure hydrocephalus 2. Wilson's disease, Huntington's disease, Parkinson's disease 3. Lewy body dementia 4. Creutzfeldt-Jakob disease



Question 27

Not answered

Marked out of 6.00

Risk factors (1)

From the attached list choose the risk factor(s) that is/are most important in the aetiology of the conditions listed below:

1. Delirium



2. Alzheimer's disease (TWO)



3. Vascular dementia (TWO)



4. Pick's disease



5. Creutzfeldt-Jakob disease

**Explanation:**

1) The most significant risk factor for delirium is the extremity of age.

2) For Alzheimer's, both age and ApoE4 are risk factors.

3) The presence of small vessel disease and diabetes are the most important independent risk factors for vascular dementia; these are more important than the age of an individual patient in vascular dementia.

4) In Pick's disease, there is positive family history in 50% of cases. It is associated with an autosomal dominant mutation in the Tau gene (chromosome 17q 21-22) with complete penetration.

5) CJD is the most common of the human prion diseases. 85% cases are spontaneous or sporadic in origin; 10% result from a genetic mutation; 5% result from iatrogenic transmission during transplant surgery of the dura, corneal grafts, and pituitary growth hormone.

Correct Answer is : 1. Age 2. Age, Apolipoprotein E4 3. Small vessel disease, Type 2 diabetes 4. Family history 5. Corneal graft surgery



Question 28

Not answered

Marked out of 12.00

Risk factors (2)

Find three most common risk factors for each of the following conditions

1. Factors associated with carer distress (Choose THREE)

2. Factors associated with depression in the elderly (Choose THREE)

3. Factors that reduce depression in carers (Choose THREE)

4. Paraphrenia in the elderly (Choose THREE)

Explanation:

1) The patients who have dementia often cause an enormous burden on carers. Higher physical dependency in the patient and history of poor relationships can worsen carer's stress.

2) Social isolation is one of the important contributory factors for depression in older subjects. Physical illness and major life events such as bereavement are other important factors contributing to depression in the elderly.

3) To reduce the incidence of depression, patient and carers should be offered adequate social support. Attendance to carer's support groups is very helpful, counselling and emotional support to carers should be encouraged.

4) Social isolation and sensory deprivation such as hearing deafness, visual impairment can contribute to psychosis in the elderly. Major life events could also precipitate psychosis in older subjects.

Correct Answer is : 1. Higher physical dependency in the patient , History of poor relationship with carer , Patient has dementia 2. Living alone , Physical illnesses, Major life events 3. Counselling and emotional support to carers, Attendance to carer's support groups, Adequate social support to both patient and carer 4. Living alone , Sensory impairment, Major life events



Question 29

Not answered

Marked out of 6.00

Choice of Antidementia Drugs (2)

Identify the most appropriate drug/drugs for each of the following situations:

1. A 76-year-old man with Alzheimer's disease developed diarrhea on taking rivastigmine. He feels that the medication is helpful and resists changing his medication. ✖

2. A 77-year-old man has been diagnosed with mixed dementia. He is disappointed to know that antidementia drugs are not indicated for the management of mixed dementia. He is keen on herbal remedies ✖

3. A 79-year-old man has been diagnosed with Alzheimer's disease. He has a significant history of cardiovascular disease and is on numerous medications. ✖

4. A 73-year-old man with Alzheimer's dementia has read about the side effects of acetylcholinesterase inhibitors, He would like to try the drug with minimal side effects. ✖

5. The family doctor (General Practitioner) of a 79-year-old man with vascular dementia seeks your advice on treating the vascular risk factors. (Choose TWO) ✖

✖

Explanation:

- 1) Rivastigmine transdermal patch is unlikely to cause any gastrointestinal side effects. Therefore, it could be used in these situations where the patient is not keen to change his medications, as he/she is benefitting out of it.
- 2) Gingko Biloba has been widely used as a cognitive enhancer. It is effective in mild to moderate Alzheimer's dementia with an advantage over placebo of around 1.4 points on ADAS-Cog.
- 3) Rivastigmine may be the safest choice in patients with cardiovascular disease given its lack of interaction potential with other drugs.
- 4) Donepezil is well tolerated and widely used. Rivastigmine and Galantamine appear more likely to cause gastrointestinal (GI) and neuropsychiatric adverse effects than does Donepezil.
- 5) Treatment of vascular risk factors aggressively is part of the management options of vascular dementia. It includes low dose aspirin, statins to reduce cholesterol, optimization of diabetic control and optimization of blood pressure.

Correct Answer is : 1. Rivastigmine transdermal patch 2. Gingko Biloba 3. Rivastigmine 4. Donepezil 5. Low dose aspirin, Statins



Question 30

Not answered

Marked out of 9.00

Clinical features of dementias

Choose the most salient clinical features for each item below:

1. Lewy body dementia (3 Answers)



2. Frontotemporal dementia (3 answers)



3. Vascular dementia (2 answers)



4. A neurological feature associated with rapidly evolving fatal dementia

**Explanation:**

1) The core features of dementia with Lewy bodies include Spontaneous motor features of Parkinsonism, progressive cognitive decline, recurrent visual hallucinations, which are typically well-formed and detailed. The supportive features include fluctuating cognition with pronounced variation in attention and alertness, neuroleptic sensitivity and history of falls.

2) Frontotemporal dementia: Features include Insidious onset and gradual progression, Early loss of personal and social awareness, Early emotional blunting, Early loss of insight. Behavioural features include early signs of disinhibition, a decline in personal hygiene & grooming, mental rigidity, inflexibility, hyperorality, stereotyped and perseverative behaviour. Speech disorder includes reduction and stereotypy of

speech, echolalia, and perseveration. Affective symptoms include Anxiety, depression, and frequent mood changes, emotional indifference. Physical signs include Incontinence, primitive reflexes, akinesia, rigidity and tremor.

3) Vascular dementia: Onset may usually follow a cerebrovascular event and is more acute in onset. The course is usually stepwise, with periods of intervening stability. Focal neurological signs & symptoms or neurological evidence of cerebrovascular disease (CVD) judged etiologically related to the disturbance.

4) Creutzfeldt-Jakob disease: The clinical picture is one of rapidly deteriorating dementia, myoclonus, cerebellar and extrapyramidal signs leading to death within a year. Patients may present with non-specific symptoms such as lethargy, depression and fatigue. Within weeks, more fulminant symptoms develop, including progressive cortical-pattern dementia, myoclonus and pyramidal and extrapyramidal signs. Myoclonus becomes prominent as the disease progresses. Patients may develop cortical blindness.

Correct Answer is : 1. Motor features of Parkinson's disease, Visual hallucinations, Fluctuating levels of consciousness 2. Insidious onset and gradual progression, Personality change and behavioural disorder, Striking loss of insight 3. Stepwise deterioration , History of transient ischemic attacks 4. Myoclonus



Question 31

Not answered

Marked out of 6.00

Antidementia drugs (1)

For the antidementia drugs given below, identify their mode of action from list above;

1. Galantamine (choose 2 answers)  

 

2. Memantine (choose 2 answers)  

 

3. Rivastigmine (choose 2 answers)  

 

Explanation:

Binding to the AChE sites may be either reversible or irreversible, and may be competitive or noncompetitive with acetylcholine. Rivastigmine is a non-selective inhibitor of both acetylcholinesterase and butyrylcholinesterase. Galantamine is a competitive drug. It is a selective and reversible inhibitor of acetylcholinesterase. It is also a nicotine modulator. Memantine is a NMDA receptor antagonist and a weak 5-HT₃ receptor antagonist.

Correct Answer is : 1. Nicotine receptor agonist, Reversible competitive inhibitor of AchEs 2. Affects glutamate transmission, NMDA receptor antagonist 3. Reversible non-competitive inhibitor of AchEs , Butrylcholinesterase inhibitor

Question 32

Not answered

Marked out of 7.00

Neuropsychometric assessments in dementia

identify 2 tests commonly used by neuropsychologists in the detailed evaluation of cognitive functions

1. Tests of executive function ✖ ✖2. Tests for practically useful episodic memory linked to daily functioning ✖3. Measurement of IQ using non-verbal methods ✖4. Useful measure of severity of dementia ✖ ✖5. Useful to differentiate vascular from other causes of dementia ✖**Explanation:**

1) Wisconsin card sorting test, Stroop test, Tower of London test, and controlled word association test are some of the tests of executive functioning.

2) Rivermead Behavioural Memory Test is used to identify everyday memory problems and monitor change over time, and assess a wider range of ability.

3) Ravens progressive matrices measures non-verbal intelligence.

4) Scales used to assess the severity of cognitive impairment include the commonly used MMSE and CAMDEX. CAMDEX stands for Cambridge examination for mental disorders of the elderly. CAMDEX incorporates the MMSE, Blessed dementia rating scale (BDRS) and the Hachinski Ischaemia score in a comprehensive inquiry from the subject and informant. BDRS is a measure of activities of daily living.

5) The Hachinski ischaemia score is helpful to differentiate between Alzheimer's and vascular dementia. A notable feature of CAMDEX is CAMCOG, which is the cognitive subscale.

Correct Answer is : 1. Wisconsin card sorting test, Stroop test 2. Rivermead Behavioural Memory test 3. Ravens progressive matrices 4. Mini-mental state examination, CAMDEX 5. Hachinski index



Question 33

Not answered

Marked out of 9.00

Aetiology of dementias

Identify 2 types of dementias for each etiological category given below

1. Degenerative cause ❌ ❌2. Intracranial ❌ ❌3. Infectious cause ❌ ❌4. Endocrine ❌ ❌5. Toxins ❌ ❌**Explanation:**

1) Degenerative dementias include Alzheimer's disease, Picks disease, Parkinson's disease, Lewy body dementia, Huntington's disease, Wilson's disease, motor neuron disease, progressive supranuclear palsy and multiple sclerosis.

- 2) Intracranial lesions causing dementia syndrome includes subdural haematoma, normal-pressure hydrocephalus, tumour and head injury.
- 3) Infectious cause include Creutzfeldt-Jakob disease, HIV-associated dementia, neurosyphilis, tuberculosis.
- 4) Endocrine causes include hypothyroidism, hyperparathyroidism, Cushing's syndrome and Addison's disease.
- 5) Toxins such as prolonged alcohol misuse, heavy metal poisoning and metabolic causes such as uraemia, hepatic encephalopathy, hypoglycemia, calcium imbalance and magnesium imbalance can also result in a dementia syndrome.

Correct Answer is : 1. Alzheimer's disease, Pick's disease 2. Subdural haematoma, Normal pressure hydrocephalus 3. HIV-associated dementia, Creutzfeldt-Jakob disease 4. hypothyroidism, Hyperparathyroidism 5. Prolonged alcohol misuse, Heavy metal poisoning



Question 34

Not answered

Marked out of 8.00

Alzheimer's disease risk factors

Identify the appropriate protective and risk factors for Alzheimer's disease

1. Proven protective factor

 ❌

2. Possible protective factors (choose two answers)

 ❌ ❌

3. Proven risk factors (choose three answers)

 ❌ ❌ ❌

4. Likely risk factors (choose two answers)

 ❌ ❌**Explanation:**

- 1) Proven, protective factors in Alzheimer's disease, include the carrier status for apolipoprotein e2 allele.
- 2) Possible protective factors include oestrogen replacement and a high level of premorbid intelligence and education. In the past smoking was considered to be possibly protective but mounting evidence suggests otherwise.
- 3) Proven risk factors include older age, Down's syndrome, carrying apolipoprotein e4 allele.
- 4) Female sex, Head injury, postmenopausal oestrogen decline are suggested as other likely risk factors.

Correct Answer is : 1. Apolipoprotein e2 allele 2. Premorbid intelligence, Oestrogen 3. Apolipoprotein e4 allele, Age, Down's syndrome 4. Head injury, Female sex



Question 35

Not answered

Marked out of 5.00

Cortical and subcortical dementias

Identify the common clinical features seen in each of the following types of dementias

1. Cortical dementias (Choose TWO)



2. Subcortical dementias (Choose THREE)

**Explanation:**

Features of cortical dementias include early aphasia, acalculia and significant memory loss in early stages. Examples of cortical dementia include Alzheimer's disease, Pick's disease, etc.

Subcortical dementias usually have no aphasia. The calculation is preserved until late, but the speed of cognitive processing is slowed early. Other features include the appearance of depressed mood, dysarthric speech, slowed motor speed & control, bowed or extended posture, apathetic personality, adventitious movements like tics, chorea, tremors and dystonia. Examples of subcortical dementias include Parkinson's disease, Huntington's disease, Wilson's disease, etc.

Correct Answer is : 1. Early aphasia, Early loss of calculation ability 2. Adventitious movements, Depressed mood, Early slowing of cognitive processing

Question 36

Not answered

Marked out of 8.00

Vascular dementias

Identify 3 appropriate clinical features for each of the different types of vascular dementias given below:

1. Multi-infarct dementia



2. Binswanger disease



3. CADASIL

**Explanation:**

1) Multi-infarct dementia has an abrupt onset with a history of multiple strokes. It is associated with a stepwise deterioration in cognitive function. Between strokes, there are periods of relative stability. There are often risk factors for cardiovascular disease.

2) Progressive small vessel disease (Binswanger's disease) is subcortical dementia with a clinical course characterised by slow intellectual decline and generalised slowing.

3) CADASIL stands for Cerebral Autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy. It is a form of vascular dementia, and the disease is transmitted as an autosomal dominant trait with high penetrance. Molecular genetic linkage studies have assigned the disease gene to the long arm of chromosome 19. Patients with this condition usually present with a recurrent stroke around later forties and there is often a history of migraine. Many patients later develop subcortical dementia and pseudo bulbar palsy. MRI shows widespread white matter changes. (Ref: http://www.rcpsych.ac.uk/pdf/semOAP_ch5.pdf (accessed July 01, 2015)).

Correct Answer is : 1. Abrupt in onset, Stepwise deterioration in cognition, Follows a number of minor ischemic episodes 2. Gradual intellectual decline , Subcortical dementia, Generalised slowing 3. Onset in their 40s, Subcortical dementia, History of migraines



Question 37

Not answered

Marked out of 6.00

Dementia care

You are seeing 3 different patients in your memory clinic. A dementia nurse at your clinic would like to seek advice and discuss options for some of his patients. Choose the most appropriate treatment options for the following presentation.

1. A 77-year-old man with MMSE score of 20/30 is unable to tolerate Donepezil. (Choose 2 answers) ✖

 ✖

2. A 82 year old woman with MMSE score of 9/30 presents with severe behavioural problems in a residential home (Choose 2 answers)

 ✖ ✖

3. A 83 year old man has been diagnosed with Alzheimer's dementia. He is on a number of different medications for physical health problems. He would like to take a memory tablet which won't interact with his other medications ✖

4. A 77-year-old man with vascular dementia. He is keen to try herbal preparations for his memory problems ✖

Explanation:

- 1) If a patient does not tolerate donepezil, they can try other drugs of the same class like galantamine and rivastigmine.
- 2) Memantine is recommended for moderate to severe dementia with behavioural problems. Risperidone is licensed for the treatment of behavioural problems of dementia in UK.
- 3) Drug-drug interaction is unlikely with rivastigmine. It is metabolised at the site of action and dose not affect hepatic cytochromes.

4) Gingko Biloba is a Chinese herbal preparation used for the treatment of mild dementia. However, the evidence is weak for its efficacy.

Correct Answer is : 1. Rivastigmine, Galantamine 2. Risperidone, Memantine 3. Rivastigmine 4. Gingko Biloba



© 2023 SPMM Course Limited, UK.

[Refund Policy](#) | [Terms & Conditions](#)

